# DWDM LAB EXPERIMENTS

**CSA-1614 · Data Warehousing and Data Mining**

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experiment 1

```
16  num1 <- as.integer(readline(prompt = "Enter the first number: "))
17  num2 <- as.integer(readline(prompt = "Enter the second number: "))
18  result <- num1 + num2
19  print(result)
```

19:14 (Top Level)

Console Terminal Background Jobs

R 4.5.2 · ~/

```
> 10
[1] 10
> 20
[1] 20
>
> source("~/.active-rstudio-document")
Enter the first number: 10
Enter the second number: 20
[1] 30
> source("~/.active-rstudio-document")
Enter the first number: 50
Enter the second number: 100
[1] 150
>
```

experiment 2

```
21  num1 <- as.integer(readline(prompt = "Enter the first number: "))
22  num2 <- as.integer(readline(prompt = "Enter the second number: "))
23  result <- num1 - num2
24  print(result)
```

24:14 (Top Level)

Console Terminal Background Jobs

R 4.5.2 · ~/

```
> source("~/.active-rstudio-document")
Enter the first number: 100
Enter the second number: 50
[1] 150
Enter the first number:
```

experiment 3

```
3.
num1 <- as.integer(readline(prompt = "Enter the first number: "))
num2 <- as.integer(readline(prompt = "Enter the second number: "))
result <- num1 * num2
print(result)
```

```
> source("~/.active-rstudio-document")
Enter the first number: 5
Enter the second number: 10
[1] 50
>
```

experiment 4

```
num1 <- as.numeric(readline(prompt = "Enter the first number: "))
num2 <- as.numeric(readline(prompt = "Enter the second number: "))

if (num2 != 0) {
  result <- num1 / num2
  print(result)
} else {
  print("Cannot divide by zero")
}
```

```
> source("~/.active-rstudio-document")
Enter the first number: 10
Enter the second number: 2
[1] 5
>
```

experiment 5

```
num <- as.integer(readline(prompt = "Enter a number: "))

if (num %% 2 == 0) {
  print("Number is even")
} else {
  print("Number is odd")
}
```

```
> source("~/.active-rstudio-document")
Enter a number: 23
[1] "Number is odd"
>
```

experiment 6

```r
student_data <- data.frame(
  name = c("Arun", "Priya", "Kiran", "Meena", "Ravi"),
  age = c(21, 22, 22, 24, 25),
  marks = c(78, 85, 91, 85, 88)
)

find_mode <- function(x) {
  values <- unique(x)
  values[which.max(tabulate(match(x, values)))]
}

print(mean(student_data$age))
print(median(student_data$age))
print(find_mode(student_data$age))
```

```
> student_data <- data.frame(
+   name = c("Arun", "Priya", "Kiran", "Meena", "Ravi"),
+   age = c(21, 22, 22, 24, 25),
+   marks = c(78, 85, 91, 85, 88)
+ )
>
> find_mode <- function(x) {
+   values <- unique(x)
+   values[which.max(tabulate(match(x, values)))]
+ }
>
> print(mean(student_data$age))
[1] 22.8
> print(median(student_data$age))
[1] 22
> print(find_mode(student_data$age))
[1] 22
>
> write.csv(student_data, "student_data.csv", row.names = FALSE)
```

experiment 7

```r
students <- data.frame(
  name = c("Anu", "Bala", "Charan", "Divya", "Rohan"),
  age = c(19, 20, 21, 22, 23),
  marks = c(76, 84, 90, 68, 95)
)

result <- summary(students)
print(result)

write.csv(students, "student_summary.csv", row.names = FALSE)

write.csv(student_data, "student_data.csv", row.names = FALSE)
```

```
> students <- data.frame(
+   name = c("Anu", "Bala", "Charan", "Divya", "Rohan"),
+   age = c(19, 20, 21, 22, 23),
+   marks = c(76, 84, 90, 68, 95)
+ )
>
> result <- summary(students)
> print(result)
     name                age        marks
 Length:5           Min.   :19   Min.   :68.0
 Class :character   1st Qu.:20   1st Qu.:76.0
 Mode  :character   Median :21   Median :84.0
                    Mean   :21   Mean   :82.6
                    3rd Qu.:22   3rd Qu.:90.0
                    Max.   :23   Max.   :95.0
>
> write.csv(students, "student_summary.csv", row.names = FALSE)
>
> write.csv(student_data, "student_data.csv", row.names = FALSE)
```

experiment 8

```
first <- as.numeric(readline("Enter the first value: "))
second <- as.numeric(readline("Enter the second value: "))
third <- as.numeric(readline("Enter the third value: "))

values <- c(first, second, third)
greatest <- max(values)

cat("The greatest number is:", greatest, "\n")
```

```
> source("~/.active-rstudio-document")
Enter the first value: 980
Enter the second value: 520
Enter the third value: 976
The greatest number is: 980
>
```

experiment 9

```
student_data <- data.frame(
  name = c("Arun", "Priya", "Kiran", "Meena", "Ravi"),
  age = c(20, 21, 22, 23, 24),
  marks = c(55, 67, 78, 89, 95)
)

result <- IQR(student_data$marks)
print(result)

write.csv(student_data, "iqr_data.csv", row.names = FALSE)
```

```
> student_data <- data.frame(
+   name = c("Arun", "Priya", "Kiran", "Meena", "Ravi"),
+   age = c(20, 21, 22, 23, 24),
+   marks = c(55, 67, 78, 89, 95)
+ )
>
> result <- IQR(student_data$marks)
> print(result)
[1] 22
>
> write.csv(student_data, "iqr_data.csv", row.names = FALSE)
```

experiment 10

```
student_data <- data.frame(
  name = c("Asha", "Bala", "Charan", "Divya", "Rohan"),
  age = c(19, 20, 21, 22, 23),
  marks = c(65, 72, 81, 88, 96)
)

result <- quantile(student_data$marks, probs = c(0, 0.25, 0.5, 0.75, 1))
print(result)

write.csv(student_data, "quantile_data.csv", row.names = FALSE)
```

```
> student_data <- data.frame(
+   name = c("Asha", "Bala", "Charan", "Divya", "Rohan"),
+   age = c(19, 20, 21, 22, 23),
+   marks = c(65, 72, 81, 88, 96)
+ )
>
> result <- quantile(student_data$marks, probs = c(0, 0.25, 0.5, 0.75, 1))
> print(result)
  0%  25%  50%  75% 100%
  65   72   81   88   96
>
> write.csv(student_data, "quantile_data.csv", row.names = FALSE)
```